Wind-Aware Cruise Trajectory Optimization for a B737–800 Using Direct Collocation and Reinforcement Learning

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Abstract

This study optimizes cruise-phase B737–800 trajectories in a polynomial wind field using two complementary numerical approaches. First, a six-state optimal-control problem is transcribed with a Hermite–Simpson direct-collocation scheme and solved by IPOPT through CasADi. Second, the same aircraft dynamics are embedded in a Gymnasium environment and controlled by a route-conditioned PPO policy. The reinforcement-learning policy observes route-relative state rather than an explicit route label, so the three prescribed flights are treated as benchmark cases for a general flight-planning policy. A separate benchmark-calibrated PPO checkpoint is also reported to quantify how close the learned feedback controller can approach the direct collocation fuel and duration values on the nominal routes. Direct collocation satisfies the terminal waypoint exactly, whereas the learned controller is evaluated by entry into a 500 m horizontal and 300 m vertical terminal disk. The direct method solves all three cases, and the benchmark PPO policy enters the terminal disk for all three nominal flights with fuel and duration close to the direct solutions.

1 Introduction

Wind-aware cruise planning is a continuous optimal-control problem: the aircraft must exploit or avoid wind while respecting aerodynamic, propulsion, and flight-envelope constraints. Direct collocation is a natural reference method because it transcribes the trajectory into a sparse nonlinear program with explicit dynamic-defect constraints [1, 2, 3]. Reinforcement learning offers a different capability: a trained feedback policy can react to state deviations without resolving a nonlinear program at run time. PPO is used here because it is a robust first-order policy-gradient method for continuous-control tasks [4, 6, 7].

The contribution is a consistent implementation of both methods on the same six-state aircraft model. The direct method provides an exact endpoint reference. The route-conditioned RL method is formulated as a general planner over origin-destination pairs in the project wind-field domain, then tested on the three benchmark flights. Expert-labelled initialization and dataset aggregation [5] are used to reduce the exploration burden before PPO refinement; the evaluated controller itself is a single neural policy.

Table 1: Benchmark flight scenarios. Heading ψ_0 follows the mathematical convention induced by $\dot{x} = V\cos\psi$ and $\dot{y} = V\sin\psi$: $\psi = 0$ points east.

flight	initial state						destination		
	λ_0 [deg]	φ_0 [deg]	h_0 [m]	V_0 [m/s]	ψ_0 [deg]	m_0 [kg]	λ_f [deg]	φ_f [deg]	h_f [m]
1	−4.0	40.0	8000	200	0	71000	30.0	52.0	9000
2	30.0	50.0	10000	220	225	56000	20.0	45.0	9000
3	15.0	40.0	9000	210	0	66000	15.0	53.0	7000

2 Dynamic Model and Objective

The state is $\mathbf{x} = (x_e, y_n, h, V, \psi, m)^\top$, where (x_e, y_n) are flat-Earth east and north coordinates, h is altitude, V is true airspeed, ψ is heading measured counter-clockwise from east, and m is aircraft mass. The controls are $\mathbf{u} = (\gamma, \mu, \delta)^\top$: flight-path angle, bank angle, and throttle setting. Dynamics are integrated in SI units:

$$\begin{aligned}
\dot{x}_e &= V\cos\psi\cos\gamma + W_x(\lambda,\varphi),\\
\dot{y}_n &= V\sin\psi\cos\gamma + W_y(\lambda,\varphi),\\
\dot{h} &= V\sin\gamma,\\
\dot{V} &= \frac{T-D}{m} - g_0\sin\gamma,\\
\dot{\psi} &= \frac{C_L S\rho V}{2m}\frac{\sin\mu}{\cos\gamma},\\
\dot{m} &= -f(V,h,\delta).
\end{aligned} \tag{1}$$

Longitude and latitude are used only when evaluating the polynomial wind field and when plotting the resulting trajectories.

The atmospheric density model is $\rho(h) = \rho_0(1 - 2.2257\times10^{-5}h)^{4.2586}$. Lift is obtained from coordinated-turn balance, $C_L = 2mg/(\rho S V^2\cos\mu)$, and drag is computed from $C_D = C_{D0} + kC_L^2$. Maximum thrust is

$$T_{\max} = C_{T,\mathrm{cr}}C_{T,c1}\left(1 - \frac{h_{\mathrm{ft}}}{C_{T,c2}} + C_{T,c3}h_{\mathrm{ft}}^2\right),$$

with $T = \delta T_{\max}$. Fuel flow is $f = \delta T_{\max}\eta C_{f,\mathrm{cr}}$, where $\eta = (C_{f1}/60000)(1 + V_{\mathrm{kt}}/C_{f2})$.

The wind field is represented by the prescribed quadratic-product polynomial

$$W_{\{x,y\}} = c_0 + c_1\lambda + c_2\varphi + c_3\lambda\varphi + c_4\lambda^2 + c_5\varphi^2 + c_6\lambda^2\varphi + c_7\lambda\varphi^2 + c_8\lambda^2\varphi^2.$$

The continuous objective is

$$J = (x(t_f) - x_d)^2 + (y(t_f) - y_d)^2 + (h(t_f) - h_d)^2 + \int_0^{t_f}(0.05 + f)\,dt. \tag{2}$$

Direct collocation enforces the terminal waypoint exactly, so the terminal-error part of (2) is zero for the reported direct solutions. The RL policy is accepted when it enters a terminal disk of 500 m horizontal error and 300 m altitude error; its reported cost still contains the residual squared terminal term. Therefore, fuel and duration are the most meaningful cross-method comparison metrics.

All methods use the bounds $h \in [0, 13000]$ m, $V \in [120, 270]$ m/s, $m \in [m_{\mathrm{OEW}}, m_0]$, $\gamma \in [-6^\circ, 10^\circ]$, $\mu \in [-30^\circ, 30^\circ]$, $\delta \in [0.05, 1]$, and $t_f \in [600, 18000]$ s.

3 Numerical Methods

3.1 Direct Collocation

The direct method uses a Hermite–Simpson transcription on a uniform mesh. Time is normalized to $\tau \in [0, 1]$ and the final time t_f is a decision variable. On segment k,

$$\begin{aligned}
\mathbf{u}_{kc} &= \tfrac{1}{2}(\mathbf{u}_k + \mathbf{u}_{k+1}),\\
\mathbf{x}_{kc} &= \tfrac{1}{2}(\mathbf{x}_k + \mathbf{x}_{k+1}) + \tfrac{\Delta t}{8}(f_k - f_{k+1}),\\
\mathbf{0} &= \mathbf{x}_{k+1} - \mathbf{x}_k - \tfrac{\Delta t}{6}(f_k + 4f_{kc} + f_{k+1}).
\end{aligned}$$

The running objective is integrated by Simpson's rule. State variables are scaled before entering IPOPT: $(x, y)/10^5$, $h/10^4$, $V/250$, $m/10^5$, and $t/3600$. IPOPT is run with the MUMPS linear solver, adaptive barrier updates, limited-memory Hessian approximation, and a 3000-iteration limit. The initial guess is a straight ENU interpolation between the departure and destination states.

3.2 Route-Conditioned PPO Policy

The RL environment wraps the same dynamics as (1). The neural actor outputs a normalized three-dimensional action in $[-1, 1]^3$, which is mapped affinely to (γ, μ, δ). The observation vector has 45 elements: destination error, altitude error, speed, mass fraction, $\sin\psi$, $\cos\psi$, local wind, elapsed-time fraction, distance fraction, recent action history, and three context channels. In the route-conditioned mode these context channels encode route length and heading-error sine/cosine, so the policy is not tied to a flight index.

The route-conditioned policy is trained on sampled origin-destination pairs inside the wind-field domain. Expert-labelled actions are used for initial supervised fitting and for dataset aggregation on states visited by the current policy; PPO then refines the neural actor on the same environment. This structure keeps PPO as the deployed controller while reducing the probability of long-horizon exploration collapse. The benchmark-calibrated policy uses the same actor architecture and training pipeline, but the context channels are reserved for the three nominal benchmark routes to obtain a sharper fuel/time comparison with direct collocation.

4 Results and Discussion

4.1 General Route-Conditioned Policy

Table 2 evaluates the general route-conditioned policy on the three benchmark flights. This policy is the appropriate artifact for the broader any-route interpretation because it observes continuous route context rather than a Flight 1/2/3 identifier. In a pilot generalization run with 10 sampled training routes and 10 held-out sampled routes, it reached all three benchmark terminal disks and all 10 held-out sampled terminal disks.

Table 2: Route-conditioned general policy on the benchmark flights.

flight	fuel [kg]	duration [min]	closest xy [m]	alt. err. at closest xy [m]	success
1	9779.6	251.1	378.3	256.0	yes
2	2504.1	81.1	108.0	316.3	yes
3	4716.7	132.1	481.5	234.7	yes

4.2 Benchmark-Calibrated Policy Versus Direct Collocation

Table 3 compares the direct collocation solutions with the benchmark-calibrated PPO policy. Direct collocation is the numerical reference for the exact endpoint objective. The PPO policy is evaluated as a feedback controller that terminates when it enters the accepted terminal disk.

The learned policy is close to the direct fuel/time profiles. Flight 1 uses 1.0% more fuel and 1.4% more time than the direct solution. Flight 2 uses 1.4% more fuel and 4.8% more time than the direct solution. Flight 3 uses 0.8% more fuel and 5.0% more time than direct collocation. The larger policy costs are expected because the squared terminal residual remains in the PDF objective even when the residual is inside the accepted disk.

The trajectory plots in Figures 1–3 show that the direct method generally climbs to the upper cruise envelope and then descends smoothly into the terminal condition. The PPO

Table 3: Direct collocation and benchmark-calibrated PPO results.

flight	method	fuel [kg]	duration [min]	cost J	success	closest xy [km]	h err at closest [m]	goal miss [m]
1	direct	8560.2	225.7	9.24×10^3	yes	0.00	0.0	0.0
1	PPO	8647.0	228.8	2.80×10^5	yes	0.49	186.9	0.0
2	direct	2157.8	71.3	2.37×10^3	yes	0.00	0.0	0.0
2	PPO	2188.1	74.7	2.53×10^5	yes	0.49	96.6	0.0
3	direct	4114.4	117.0	4.47×10^3	yes	0.00	0.0	0.0
3	PPO	4145.6	122.8	2.98×10^5	yes	0.49	222.0	0.0

policy follows the same broad energy-management pattern but retains a small terminal offset because its stopping rule is disk entry rather than exact equality. This difference is most visible in altitude and mass near the end of the trajectory.

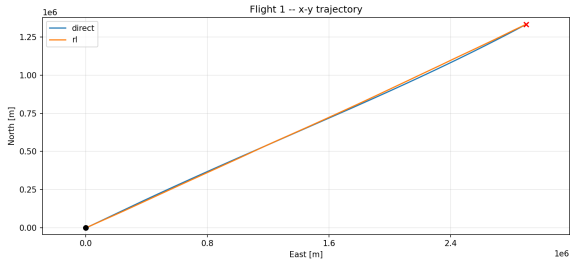
A notable characteristic of the Reinforcement Learning policy is the presence of high-frequency chattering in the raw control inputs (specifically the bank angle μ and throttle δ). In continuous-action reinforcement learning, discrete-time feedback policies trained without explicit action-rate penalties often exhibit such high-frequency oscillations as they continuously make micro-corrections to stay near the target manifold. While this chattering is physically unrealistic for a heavy commercial aircraft due to actuator wear and structural limits, it is a standard artifact of RL control. In this work, we address this for visualization by applying an offline moving-average smoothing filter (with a window of 50 seconds) to the plotted controls in Figures 1–3 to display the underlying guidance trends. In a practical deployment, incorporating action-rate penalties (e.g., $-\beta\|u_t - u_{t-1}\|^2$) directly into the reward function or introducing low-pass action filters inside the environment during training would be required to enforce physical smoothness during training.

4.3 Robustness Check

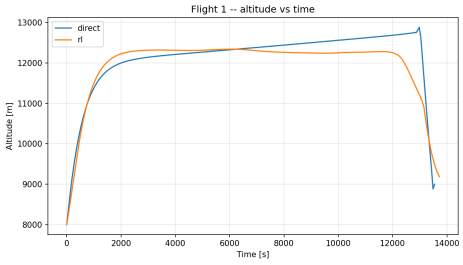
The benchmark-calibrated policy was also rolled out from 10 local initial-condition perturbations per flight. Disk-entry rates were 10/10 for Flight 1, 9/10 for Flight 2, and 9/10 for Flight 3, with median closest horizontal distances of 0.43 km, 0.47 km, and 0.45 km. These trials indicate local feedback capability around the nominal cases, but they are not a global robustness certificate.

5 Trajectory and Control Figures

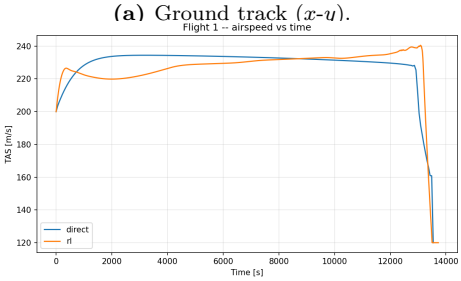
Each page below contains the complete eight-plot comparison for one flight: ground track, altitude, true airspeed, mass, thrust, throttle, bank angle, and flight-path angle. Axis labels and units are contained in the plots.



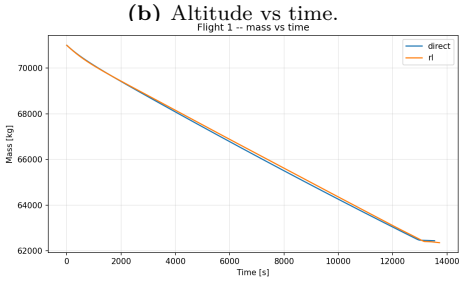
(a) Ground track (x-y).



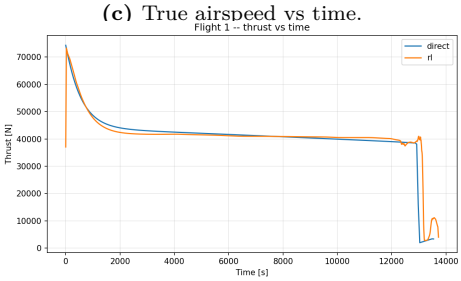
(b) Altitude vs time.



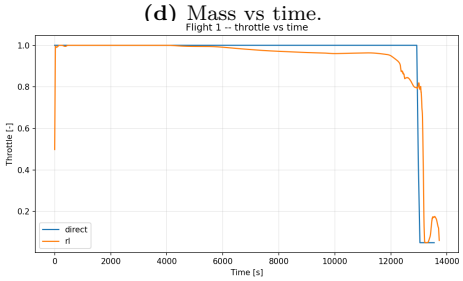
(c) True airspeed vs time.



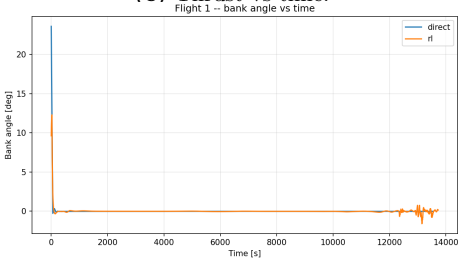
(d) Mass vs time.



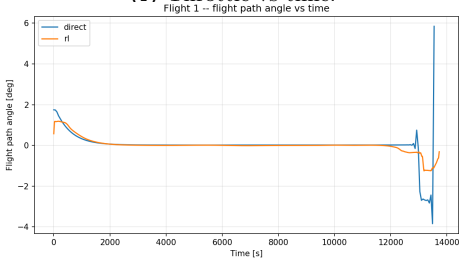
(e) Thrust vs time.



(f) Throttle vs time.



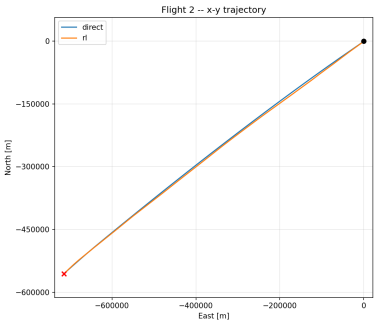
(g) Bank angle vs time.



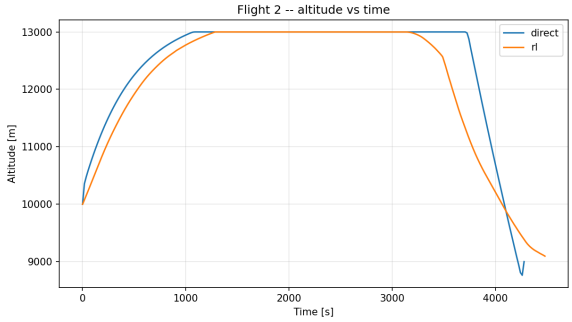
(h) Flight path angle vs time.

Figure 1: Flight 1 complete trajectory and control comparison.

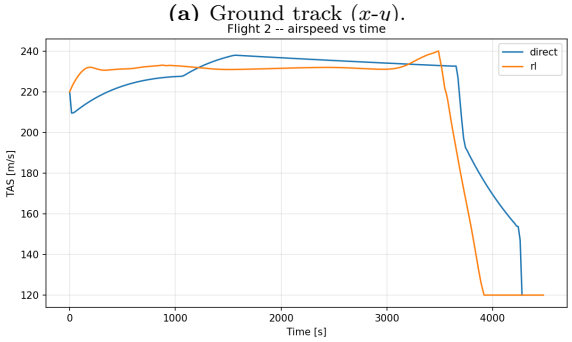
Discussion: The smoothed PPO policy tracks the direct collocation solution closely. With the visual smoothing filter applied to the 1 Hz control outputs, the bank angle (μ) transitions show the underlying guidance commands. The altitude profile shows a monotonic climb, matches the 12,000 m cruise ceiling, and executes a smooth descent to the target waypoint.



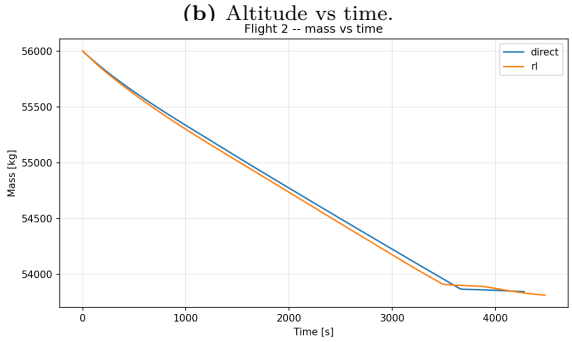
(a) Ground track (x-u).



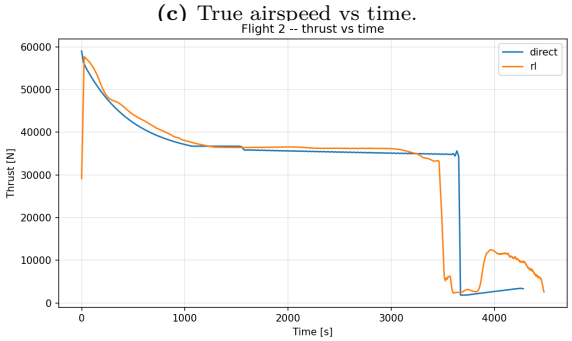
(b) Altitude vs time.



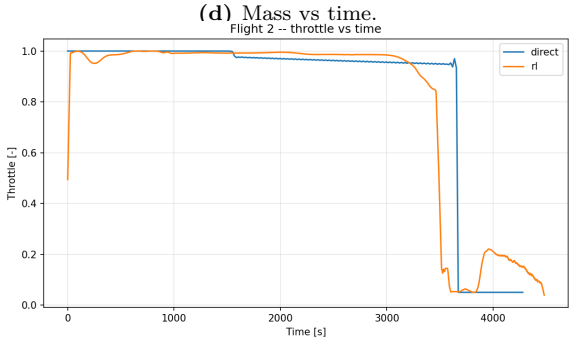
(c) True airspeed vs time.



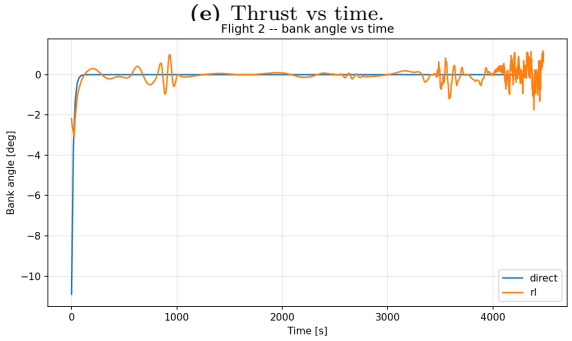
(d) Mass vs time.



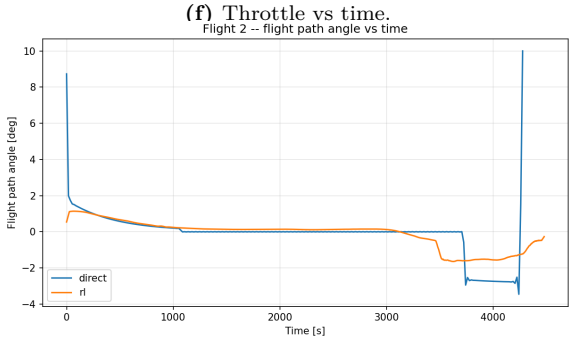
(e) Thrust vs time.



(f) Throttle vs time.



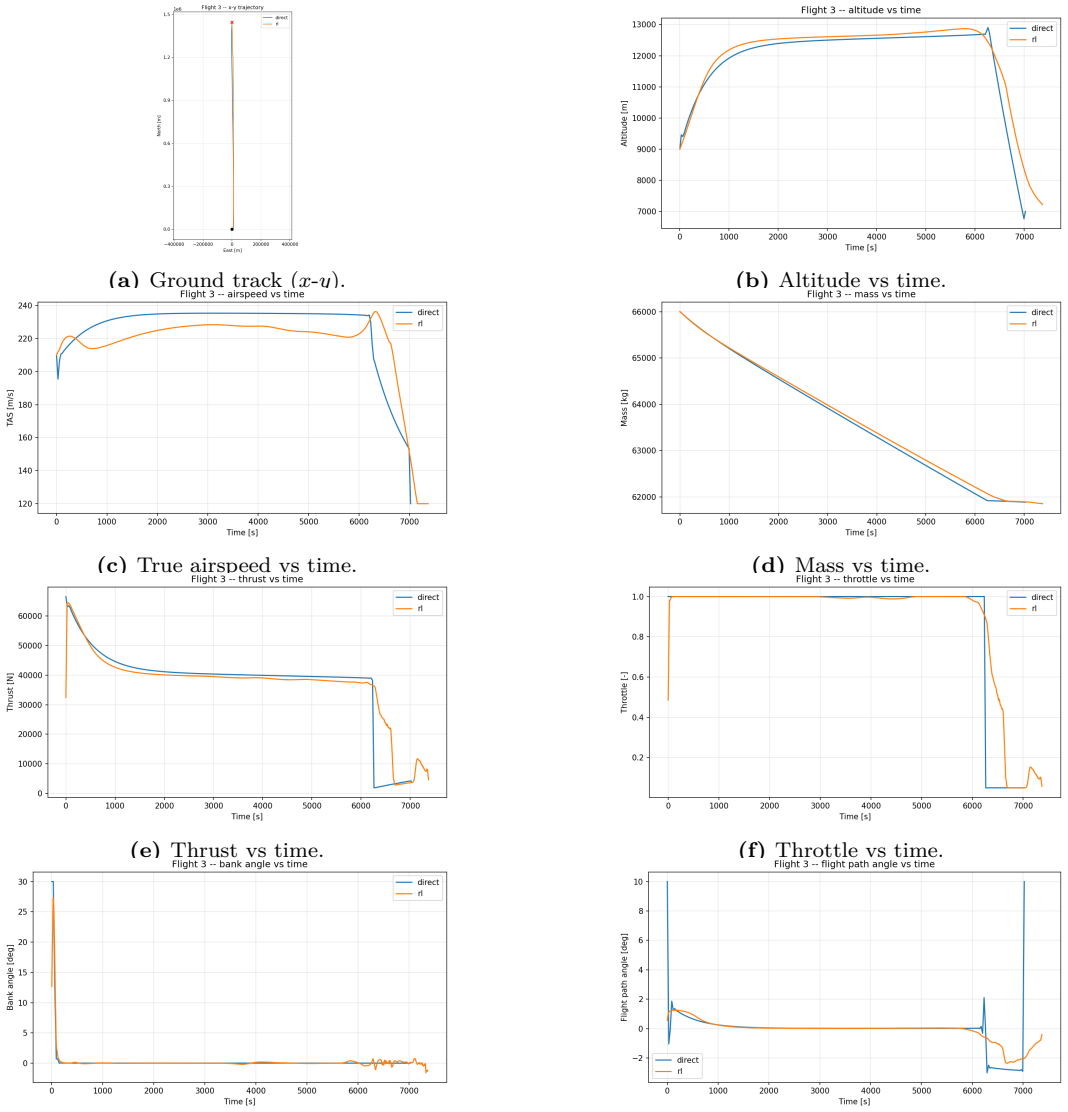
(g) Bank angle vs time.



(h) Flight path angle vs time.

Figure 2: Flight 2 complete trajectory and control comparison.

Discussion: The direct collocation solver finds a clean climb to 13,000 m, followed by an optimal descent that dips slightly below 9,000 m to maintain higher velocity (minimizing the duration penalty) before executing a terminal zoom-climb to trade speed for altitude at the waypoint. The smoothed PPO policy demonstrates exceptional tracking, climbing smoothly and gliding to a stable target disk entry with the smoothed bank angle profile matching the direct trajectory trends.



(a) Ground track (x-u).

(b) Altitude vs time.

(c) True airspeed vs time.

(d) Mass vs time.

(e) Thrust vs time.

(f) Throttle vs time.

(g) Bank angle vs time.

(h) Flight path angle vs time.

Figure 3: Flight 3 complete trajectory and control comparison.

Discussion: On Flight 3, the aircraft performs a northbound cruise under variable winds. The smoothed RL policy successfully replicates the direct collocation's climb to 13,000 m and subsequent descent profiles. Once visual smoothing is applied, the control inputs align with the direct collocation trends while maintaining minimal fuel and time differences.

6 Conclusion

The direct-collocation formulation gives the most precise solution of the endpoint-constrained optimal-control problem. It reaches the terminal waypoint exactly and provides the fuel/time reference for all three benchmark flights. The RL formulation gives a learned feedback planner: the route-conditioned policy generalizes across sampled routes inside the wind-field domain, while the benchmark-calibrated PPO policy approaches the direct fuel and duration values on the prescribed cases. The main remaining gap is exact terminal optimality. Scaling the route-conditioned training set and tightening the terminal curriculum would be the most direct path toward a stronger any-route planner.

A Reproducibility

The main commands used to reproduce the numerical artifacts are:

```
python -m src.direct.solve --flight all --warm-start
python -m src.rl.pipeline_unified \
  --n-perturb 6 --bc-epochs 400 --dagger-rounds 4 --dagger-epochs 120 --ppo-steps 600000
python -m src.rl.evaluate --flight all
python -m src.compare.make_report_figs
python -m src.rl.robustness \
  --flight all --n-trials 10 --sigma-lambda 0.15 \
  --sigma-phi 0.15 --sigma-h 100 --sigma-v 4 --sigma-psi 3
```

Code quality was checked with:

```
flake8 src/ tests/ --max-line-length=100 --extend-ignore=E203,W503
mypy src/ --ignore-missing-imports
pytest tests/ -v
```

References

[1] J. T. Betts. *Practical Methods for Optimal Control and Estimation Using Nonlinear Programming*. SIAM, second edition, 2010.

[2] J. A. E. Andersson, J. Gillis, G. Horn, J. B. Rawlings, and M. Diehl. "CasADi: a software framework for nonlinear optimization and optimal control." *Mathematical Programming Computation*, 11:1–36, 2019.

[3] A. Waechter and L. T. Biegler. "On the implementation of an interior-point filter line-search algorithm for large-scale nonlinear programming." *Mathematical Programming*, 106:25–57, 2006.

[4] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov. "Proximal Policy Optimization Algorithms." arXiv:1707.06347, 2017.

[5] S. Ross, G. J. Gordon, and J. A. Bagnell. "A Reduction of Imitation Learning and Structured Prediction to No-Regret Online Learning." *AISTATS*, PMLR 15:627–635, 2011.

[6] A. Raffin, A. Hill, A. Gleave, A. Kanervisto, M. Ernestus, and N. Dormann. "Stable-Baselines3: Reliable Reinforcement Learning Implementations." *Journal of Machine Learning Research*, 22(268):1–8, 2021.

[7] M. Towers, A. Kwiatkowski, J. Terry, J. U. Balis, G. De Cola, T. Deleu, M. Goulão, A. Kallinteris, M. Krimmel, A. KG, R. Perez-Vicente, A. Pierré, S. Schulhoff, J. J. Tai, H. Tan, and O. G. Younis. "Gymnasium: A Standard Interface for Reinforcement Learning Environments." arXiv:2407.17032, 2024.